Code Coverage Report

Project: Customer Churn Prediction (Team Hackforge)
Course: Software Engineering Mini Project, Deliverable Part 3
Date of test run: 30 October 2026
Generated by: `python tools/generate_test_reports.py` from a single run of the automated test suite

1. Method

Statement coverage was measured with coverage.py (through pytest-cov) while the full automated test suite ran. It counts the share of executable statements in each application module that ran at least once. Test files, `conftest.py`, `traceability.py` and the `tools/` folder are excluded so the figures describe the application code only.

2. Overall Result

Statements	Covered	Missed	Statement coverage
1444	1286	158	89.1%

3. Coverage by Module

Module	Statements	Covered	Missed	Coverage
`01_churn_prediction_xgboost.py`	72	71	1	98.6%
`02_paradigm_benchmark.py`	103	93	10	90.3%
`app.py`	513	396	117	77.2%
`classical_models.py`	39	39	0	100.0%
`customer_presets.py`	8	8	0	100.0%
`dml_model.py`	154	150	4	97.4%
`prediction_pipeline.py`	363	337	26	92.8%
`qml_model.py`	34	34	0	100.0%
`training_common.py`	158	158	0	100.0%

4. Statements Not Executed

Module	Lines
`01_churn_prediction_xgboost.py`	128
`02_paradigm_benchmark.py`	213-221, 225
`app.py`	25-26, 343-346, 382, 503-504, 574-575, 590-592, 600, 684-686, 700-701, 725-727, 743-745, 761-763, 779-781, 886, 963, 967, 1080-1081, 1125, 1139-1140, 1155, 1157-1159, 1166-1167, 1172-1173, 1175-1176, 1182-1189, 1191-1197, 1199-1204, 1212-1213, 1215-1219, 1222-1225, 1227-1235, 1237-1241, 1244-1247, 1251, 1259, 1261-1263, 1271-1272, 1281-1287, 1365, 1397-1398

Module	Lines
classical_models.py	none
customer_presets.py	none
dml_model.py	114, 205-207
prediction_pipeline.py	163-164, 222, 226-227, 235, 341, 361, 399, 458, 502-504, 614, 621, 657-658, 746-747, 807-808, 839, 844, 875-877
qml_model.py	none
training_common.py	none

Most of the remaining statements in app.py are Streamlit page sections that are only reached through browser interactions such as file uploads and download buttons; these flows are shown working in the Sprint 2 demonstration video.

5. Evidence

Machine-readable results are in raw/coverage.xml (Cobertura format) and raw/coverage.json.